Hemlo

HEMLO · SYSTEM DESIGN

Hemlo Ecosystem Design

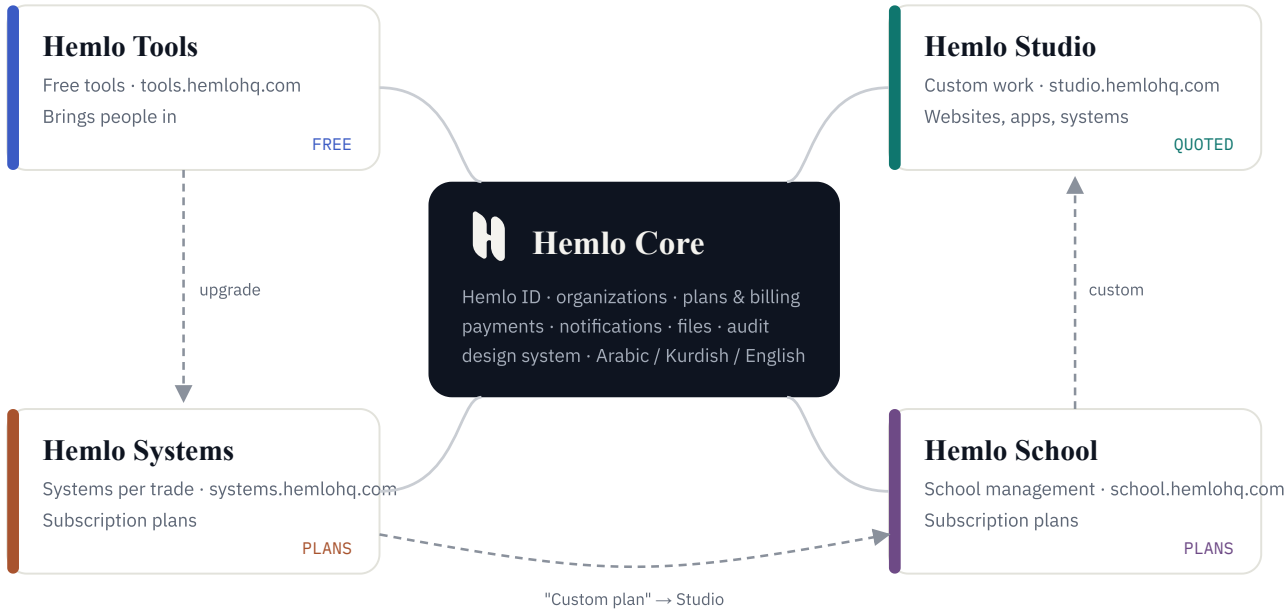
One brand, four branches: what each one does, how they connect, how they make money, and the system that runs them at $0 until the first subscriber

حلولٌ بإتقان.

Version 1 · 27 September 2026 · Builds on the brand book, tech-stack-plan.md and the $0 launch decision.

1. The ecosystem at a glance

Hemlo is the parent brand. Under it sit **four branches**, each with its own job, audience, address and colour, all sharing one account system, one design language and one platform.



BRANCH	ARABIC	ADDRESS	COLOUR	JOB IN THE BUSINESS	HOW IT EARNS
Hemlo (parent)	هیملو	`hemlohq.com`	Ink + Brass	The hub: brand, trust, routes people to branches	Doesn't sell anything itself
Hemlo Tools	هیملو · الأدوات	`tools.hemlohq.com`	Cobalt `#3B5BC4`	Free, useful tools that bring businesses in and show which branch they need	**Free forever**; leads to paid branches
Hemlo Studio	هیملو · ستودیو	`studio.hemlohq.com`	Teal `#0F766E`	Custom websites, apps and systems, including custom plans for other branches	**Quoted per project** + maintenance retainers
Hemlo Systems	هیملو · الأنظمة	`systems.hemlohq.com`	Rust `#A8532E`	Ready-made systems, one edition per trade; first edition: repair workshops (inventory, job cards, warranty)	**Monthly / annual plans**
Hemlo School	هیملو · المدارس	`school.hemlohq.com`	Plum `#6E4A86`	Complete management for private schools and kindergartens	**Plans by student count**

How the branches feed each other

1. **Tools brings people in.** A workshop owner uses the free receipt generator, and a school accountant uses the fee-schedule tool.
2. **Every tool has a natural next step** into a paid branch: "Save your receipts and track stock → try Hemlo Systems."
3. **Every paid branch has a "Custom" plan** that sends the customer to **Studio** with their details already filled in.
4. **Studio earns money from day one** (custom work needs no servers). That revenue funds building the products.
5. **One Hemlo account (Hemlo ID)** works everywhere, so a customer who joins one branch can add another in two clicks.

2. The brand rules for branches

- **Naming:** "Hemlo" + one descriptor word. In Arabic the descriptor is a label after a middle dot (`هيملو · الأنظمة`), so it never reads like a company name.
- **Logo:** the founder's **H** with "Hemlo" in Poppins SemiBold. In a branch logo **the branch name goes under "Hemlo" in Poppins Light**, and the H takes the branch colour (the parent keeps ink or brass). The H alone, on a tile in the branch colour, is each branch's app icon. Files: `Visual Identities/Hemlo/logo/svg/branches/` .
- **One design system (`@hemlo/ui`)** for every branch. A branch changes only its accent colour, its icons and its content.
- **Tools always carries a "Free" badge,** so it's never mistaken for a paid product.
- **Custom work always says Studio.** "Custom plan" buttons in Systems and School open a Studio brief, never a hidden sales process.

BRANCH	H COLOUR	BRANCH NAME	LOGO FILES
Hemlo	Ink (brass for brand moments)	none	`hemlo-lockup.svg`
Tools	Cobalt `#3B5BC4`	"Tools" under "Hemlo", Poppins Light · *Free* badge	`branches/hemlo-tools*.svg`
Studio	Teal `#0F766E`	"Studio" under "Hemlo", Poppins Light	`branches/hemlo-studio*.svg`
Systems	Rust `#A8532E`	"Systems" under "Hemlo", Poppins Light	`branches/hemlo-systems*.svg`
School	Plum `#6E4A86`	"School" under "Hemlo", Poppins Light	`branches/hemlo-school*.svg`

3. Hemlo Tools: free, forever (`tools.hemlohq.com`)

Purpose

Genuinely useful, Iraq-specific tools that cost **$0 to run**, because they work entirely in the visitor's browser. They bring businesses to Hemlo, build trust, rank in search, and tell you which paid branch attracts the most interest.

Rules for every tool

- It must solve a **real, recurring Iraqi task** in under two minutes.
- It **runs in the browser**: no account, no server, no cost. Accounts are optional, only to *save* results.
- Arabic first, with English and Kurdish where relevant. Documents come out as **clean PDFs** (browser print with correct Arabic shaping).
- Every result carries a small "Made with Hemlo Tools" line, and a **next step** to the right branch.
- Estimates are clearly labelled as estimates. No tool gives legal or tax advice.

The tools

Launch set (build first):

TOOL	ARABIC	WHAT IT DOES	WHY IRAQIS NEED IT	LEADS TO
Invoice & quotation maker	صانع الفواتير وعروض الأسعار	Bilingual invoices and quotes in IQD/USD, with logo, stamp area and QR; exports to PDF	Most small businesses still write these by hand or in Word	Systems
Amount in words (تفقيط)	تفقيط المبالغ	Converts numbers to written Arabic and English words ("مليون ومئتان وخمسون ألف دينار عراقي فقط لا غير")	Required on cheques, invoices and contracts; people type it by hand and make mistakes	Systems, School
Repair receipt & warranty card	وصل التصليح وكارت الضمان	Job receipt with the customer's device and fault, warranty period and QR; prints on A5 or thermal paper	Systems lose track of what they received and what's under warranty	Systems
School fee instalment planner	مخطط أقساط المدرسة	Splits annual fees into instalments, with due dates, a family statement and receipts	School accountants rebuild this every year in Excel	School
Import landed-cost estimator	حاسبة كلفة الاستيراد	Estimates duties, tax deposits, shipping and fees for a shipment in IQD/USD	New 2026 customs rules made import costs hard to predict	Studio (custom import systems)

Second wave:

TOOL	ARABIC	WHAT IT DOES	LEADS TO
Salary & social-security calculator	حاسبة الرواتب والضمان	Employee net pay and Daman contributions (12% employer / 5% employee)	Studio (payroll), Systems
Stock count sheet	ورقة جرد المخزون	A printable or phone checklist for counting stock, with a totals summary	Systems
Class timetable builder	منظم الجدول الدراسي	Drag-and-drop weekly timetable per class, exported to PDF	School
Report card maker	صانع الشهادات المدرسية	Bilingual report cards from a simple grade table	School
Hijri ↔ Gregorian date converter	محول التاريخ	Fast, with printable calendars	Everyone (search traffic)
QR maker for businesses	صانع رمز QR	WhatsApp, location and menu QR codes with a clean frame	Systems
Project cost estimator	حاسبة كلفة المشروع البرمجي	Choose what you need (site, app, system) and get a price range	**Studio** (becomes a real brief)

Growth mechanics

- Every tool page is a **search landing page** in Arabic ("برنامج فواتير مجاني", "تفقيط المبالغ", "حاسبة كلفة الاستيراد العراق").
- Results are **shareable** (PDF or link), which spreads the brand through WhatsApp.
- An optional free **Hemlo ID** saves history, templates and your logo. That's the bridge to paid branches.
- Track which tools convert to which branch; build more of what converts.

4. Hemlo Studio: custom work (`studio.hemlohq.com`)

Purpose

Custom websites, web apps, mobile apps and business systems for organizations whose needs don't fit a standard plan, **including "Custom" plans of Systems and School**. Studio is also the fastest route to revenue: it needs no infrastructure and pays for building the products.

Services

SERVICE	ARABIC	TYPICAL SCOPE
Business websites	مواقع الشركات	Bilingual site, pages, contact/WhatsApp, basic search optimisation
Online stores & catalogues	المتاجر الإلكترونية	Catalogue, orders, COD, FIB/ZainCash payment links
Web applications	تطبيقات الويب	Portals, dashboards, booking, internal tools
Mobile apps	تطبيقات الموبايل	Built once for iOS and Android
Custom business systems	الأنظمة المخصصة	ERP-style modules, approvals, reporting, integrations
Custom plans for Systems / School	الخطط المخصصة	Extra modules, integrations, migration, on-site training on top of a Hemlo product
Care plans (maintenance)	خطط الرعاية	Hosting, updates, backups, small changes, support hours

Flexible pricing (how quotes are built)

Studio prices are **quoted, not listed**, using one transparent formula so they stay consistent:

> **Quote = base package + modules + complexity + integrations + delivery speed**, then an optional **monthly care plan**.

PART	HOW IT WORKS
Base package	A fixed starting price per service type (site, store, web app, mobile app, system)
Modules	Each extra feature has a price (e.g., booking, payments, multi-language, admin roles, reports)
Complexity	×1.0 simple · ×1.3 moderate · ×1.6 complex (data migration, approvals, heavy logic)
Integrations	Priced per integration (FIB/ZainCash, WhatsApp API, SMS, accounting export)
Delivery speed	Standard, or rush (+25%)
Payment terms	50% deposit, 30% at the midpoint milestone, 20% on delivery; larger projects by milestone

PART	HOW IT WORKS
Care plan	Monthly retainer tiers (Basic / Business / Priority) by included hours and response time

Starting prices are **hypotheses to calibrate after the first five quotes**. Keep a quote log (brief → price → hours actually spent) and adjust the base prices every quarter.

The Studio website and portal

- **Public site:** services, portfolio (starting with Hemlo's own branches and brand kit), the process, and a **"Build your quote" configurator**. The configurator is the Project cost estimator from Tools, which turns into a brief.
- **Client portal (after sign-in):** project status and milestones, shared files, change requests, invoices and payment links, and care-plan hours.
- **The process:** brief → estimate → discovery call → fixed quote → deposit → milestones → delivery → care plan.

Guardrails

- Studio must **not** swallow product time. Cap Studio at a fixed share of the week (e.g., 50%) and review monthly.
- Reusable work from Studio projects feeds the shared platform (`@hemlo/ui` , core packages). Customer-specific code stays in the customer's project.
- Every project has a written scope. Changes go through change requests with a price.

5. Hemlo Systems: one core, an edition per trade (`systems.hemlohq.com`)

Editions

One shared core (customers and WhatsApp, stock, checkout/POS and receipts, payments, staff and permissions, reports, offline sync) with an **edition** per kind of business on top. Editions are built **one at a time**, and the next starts only when the current one has about 10 paying customers.

ORDER	EDITION	ADDS ON TOP OF THE CORE	STATUS
1	**Workshops** (repair, installation, service)	Job cards, device status, "ready" WhatsApp, parts per job, warranty	First
2	Appointment businesses (salons, clinics, car services)	Appointments and services; reuses most of the workshop screens	Planned; check Iraqi competitors first
3	Cafés and restaurants	Tables, kitchen orders, menu	Planned; crowded market (QR Iraq, Ejaf, RA8M)
4	Stores	Barcode, fast checkout, suppliers	Planned; crowded market (RA8M, Geno)
5	Hotels	Rooms, bookings calendar, check-in/out, housekeeping	Last; custom Studio projects until then

POS is not a separate product: it is the core checkout, fitted to each edition. The rest of this section describes the first edition.

Positioning

Generic inventory and point-of-sale software is crowded in Iraq (Geno, Giturn, Al-Ameen, Daftra, Invent ERP and others). Systems wins by being **built for businesses that repair, install and service**: mechanics, AC and refrigeration, phone and electronics repair, generators, solar installers, carpentry and metal shops. They need **stock + jobs + warranty** in one place, which generic tools don't do well.

What's in it

MVP (first release):

- **Items & stock:** parts and products, barcodes/QR, units, cost and sale price in IQD/USD, minimum stock alerts
- **Stock movements:** receive, sell, use in a job, adjust, transfer; a full history per item
- **Job cards:** customer, device or vehicle, reported fault, parts used (deducted from stock), labour, status, pickup
- **Warranty:** a warranty period per job or part, a QR warranty card, and a warranty lookup
- **Sales & receipts:** quick sale, bilingual receipt, cash / FIB / ZainCash / credit
- **Customers & credit:** balances, WhatsApp statement links

- **Reports:** stock value, low stock, jobs by status, sales by day

Later:

- Purchase orders and suppliers
- Multiple branches and stock transfers
- Technician app with offline use
- Thermal printer support
- Accounting export
- A public API

Plans (starting hypotheses in IQD; validate before launch)

PLAN	FOR	INCLUDES	PRICE / MONTH
Free trial	Everyone	Full Pro plan for 30 days	0
Starter	One-person shop	1 location · 2 users · up to 1,000 items · jobs · receipts	≈ 25,000
Pro	Growing workshop	1 location · 5 users · unlimited items · warranty · WhatsApp statements · reports	≈ 50,000
Business	Several branches	Up to 3 locations · 15 users · transfers · purchase orders · priority support	≈ 100,000
Custom	Special needs	Anything beyond Business	→ **Hemlo Studio** (quoted)

Annual billing gets two months free. Anchor for comparison: Geno Cloud ERP charges about $9 per user per month with a minimum of 5 users.

6. Hemlo School: school management (`school.hemlohq.com`)

Positioning

School management systems exist in the region. School wins by being **Iraq-native and parent-friendly**:

- Arabic-first, with Kurdish for the Kurdistan Region.
- Fee instalments and reminders built around how Iraqi families actually pay.
- Parents get **WhatsApp links**, not another app to install.
- Report cards follow the Iraqi grading conventions each school uses (configurable).

What's in it

MVP (first release):

- **Students & families:** profiles, guardians, documents, siblings linked
- **Admissions:** applications, interviews, acceptance, enrolment
- **Classes & sections:** grades, sections, class teachers, subjects
- **Fees:** fee plans, instalments, discounts (siblings, staff), receipts, overdue lists, **WhatsApp reminders with statement links**, and FIB/ZainCash/cash payments
- **Attendance:** daily by class, from a teacher's phone (works offline), with absence alerts to parents
- **Grades & report cards:** configurable grading scales, bilingual report cards as PDFs
- **Announcements:** to classes, grades or the whole school, delivered as links

Later:

- Timetables
- Staff and payroll
- Exams and question banks
- Transport routes and bus fees
- Library
- A parent portal with a family login
- Ministry reports

Plans (starting hypotheses in IQD; validate with schools)

PLAN	FOR	INCLUDES	PRICE
Free trial	Every school	Full Pro plan for 30 days	0
Kindergarten	Up to 150 students	Students, fees, attendance, announcements	≈ 100,000 / month
Pro	Up to 600 students	Everything in the MVP	≈ 200,000 / month

PLAN	FOR	INCLUDES	PRICE
Campus	Up to 1,500 students, several campuses	Pro + multi-campus + priority support	≈ 350,000 / month
Custom	Larger groups or special needs	Integrations, migration, on-site training	→ **Hemlo Studio** (quoted)

Annual plans are paid per term or per year (schools budget by year). Sell before the school year starts.

7. The shared platform: Hemlo Core

Every branch uses the same core services. This is what makes four branches possible for one person.

CORE SERVICE	WHAT IT DOES	USED BY
Hemlo ID (`account.hemlohq.com`)	One login for all branches. Phone or email codes, organizations, members, roles	All
Organizations & roles	A business, school or client, with members, roles (owner, admin, staff, teacher, parent) and branch access	All
Plans, subscriptions & entitlements	Which organization has which branch and plan, trials, limits (users, items, students)	Systems, School, Studio care plans
Billing & payments	Invoices in IQD/USD; FIB, ZainCash, bank transfer and cash (recorded); receipts; reminders	All paid branches
Notifications	WhatsApp links / Cloud API, SMS codes, email, in-app	All
Files	Uploads (documents, photos, logos) to R2 with signed links	All
Audit log	Who changed what and when, for money and records	All
Documents engine	Bilingual PDF templates (invoices, receipts, report cards, warranty cards)	Tools, Systems, School, Studio
Design system (`@hemlo/ui`)	Components, tokens, right-to-left, dark mode, branch accents	All
i18n	Arabic, Kurdish, English; IQD formatting; amount in words; Hijri/Gregorian	All
Leads & briefs	"Custom plan" requests and tool-to-branch sign-ups, handed to Studio or sales	Tools, Studio, all "Custom" buttons

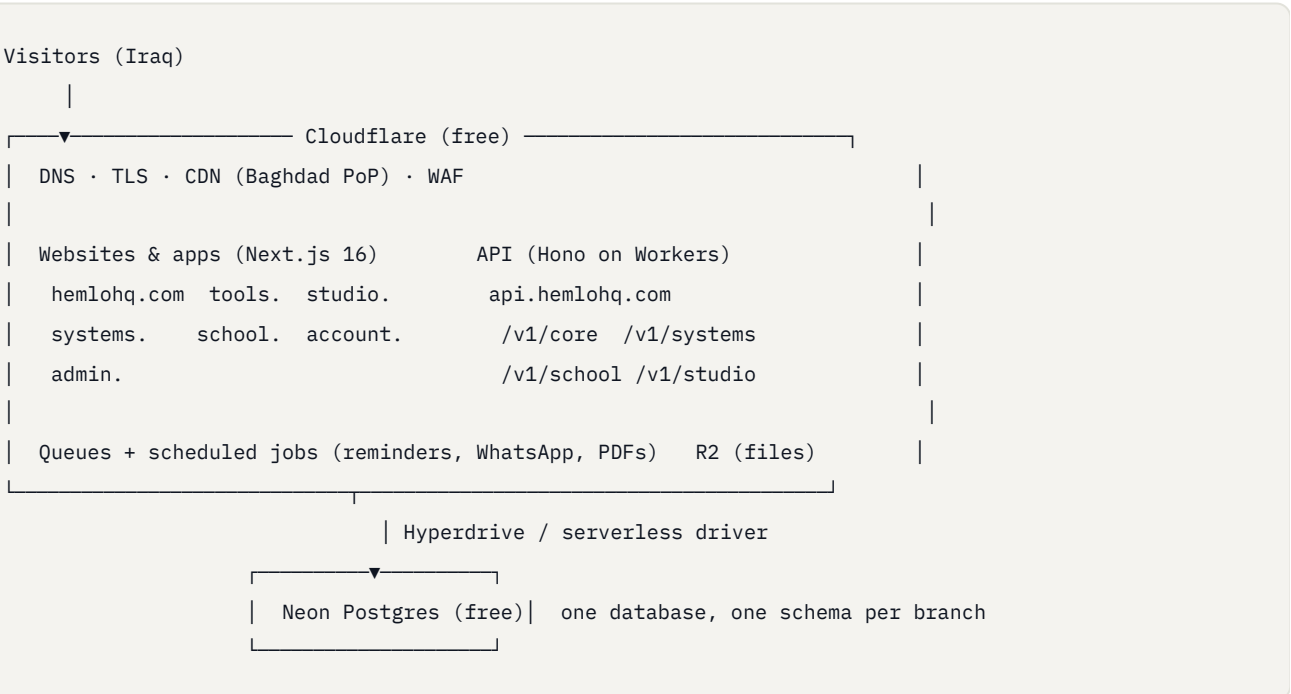
8. System architecture

Addresses

ADDRESS	WHAT LIVES THERE
`hemlohq.com`	The parent website (hub), in Arabic and English
`tools.hemlohq.com`	All free tools (static and browser-only)
`studio.hemlohq.com`	The Studio website + the client portal
`systems.hemlohq.com`	The Systems website + the app (`/app`)
`school.hemlohq.com`	The School website + the app (`/app`)
`account.hemlohq.com`	Hemlo ID: login, organizations, billing, plans
`api.hemlohq.com`	The one shared API (all branches)
`admin.hemlohq.com`	Your internal back office (customers, plans, payments, support)
`status.hemlohq.com`	A public status page (later)

The login cookie is set on `.hemlohq.com`, so signing in once works on every branch.

How it runs at $0



LAYER	TECHNOLOGY	WHY
Websites & apps	**Next.js 16** on Cloudflare (OpenNext)	One framework for every site and app; free hosting
Free tools	Next.js static pages, logic in the browser	Zero server cost, works even at scale
API	**Hono** on Cloudflare Workers	Runs free on Workers now, and on a normal Node server later with no rewrite
Database	**PostgreSQL (Neon free)** + **Drizzle**	Real transactions for money; the same database when you upgrade
Login	**Better Auth** (organizations, phone/email codes)	Your own users in your own database. Codes instead of passwords keep CPU use inside the free limit.
Files	**Cloudflare R2**	10 GB free, no download fees
Background work	**Cloudflare Queues + Cron**	Reminders, messages and scheduled reports inside the free limits
Documents	HTML templates → PDF (browser print for Tools; a rendering service later for apps)	Correct Arabic shaping
Design	**Tailwind v4** + **React Aria** + brand tokens	One consistent Hemlo look, right-to-left by default

One database, a schema per branch

SCHEMA	HOLDS
`core`	users, organizations, members, roles, plans, subscriptions, invoices, payments, files, audit, notifications, leads
`systems`	items, stock movements, locations, job cards, warranties, sales, customers, credit
`school`	students, guardians, classes, sections, subjects, fee plans, instalments, attendance, grades, report cards
`studio`	clients, briefs, quotes, projects, milestones, change requests, care plans, time logs
`tools`	(optional) saved results and templates for signed-in users

- Every business table has `org_id`, and **row-level security** keeps organizations apart.
- Branches never read each other's tables directly. They go through Core (for example, a Systems "Custom plan" creates a **lead** in Core, which Studio picks up).
- Because each branch has its own schema, API routes and app, any branch can later **move to its own service or database** without touching the others.

Codebase (one repository)

```
hemlo/
├─ apps/
│  ├─ hub/          hemlohq.com
│  ├─ tools/        tools.hemlohq.com     (static, browser-only)
│  ├─ studio/       studio.hemlohq.com    (site + client portal)
│  ├─ systems/      systems.hemlohq.com   (site + /app)
│  ├─ school/       school.hemlohq.com    (site + /app)
│  ├─ account/      account.hemlohq.com   (Hemlo ID, billing)
│  ├─ admin/        admin.hemlohq.com     (internal)
│  └─ api/          api.hemlohq.com       (Hono: /v1/core, /v1/systems, /v1/school, /v1/studio)
└─ packages/
   ├─ ui/           design system + branch themes
   ├─ core/         money (IQD/USD), ids, dates (Hijri/Gregorian), amount-in-words, Arabic search
   ├─ db/           Drizzle schemas (core, systems, school, studio), migrations, security rules
   ├─ auth/         Better Auth setup
   ├─ billing/      plans, entitlements, invoices, payment adapters (FIB, ZainCash, cash)
   ├─ notify/       WhatsApp, SMS, email
   ├─ documents/    bilingual PDF templates
   ├─ i18n/         ar, ckb, en
   └─ branch-*/     each branch's business rules (systems, school, studio)
```

Plans and entitlements (how one system sells four branches)

- An **organization** holds **subscriptions**, one per branch (e.g., Systems Pro + a Studio care plan).
- Each plan defines **entitlements**: limits (users, items, students, locations) and features (warranty, transfers, multi-campus).
- The app checks entitlements, never plan names. Changing a price or plan never requires code changes.
- **Trials** are subscriptions with an end date. Downgrades keep data read-only instead of deleting it.
- Invoices are generated monthly or annually and paid by FIB/ZainCash links or recorded as cash or bank transfer. Unpaid invoices get WhatsApp reminders, then a grace period, then read-only mode.

The "Custom plan" flow (Systems or School → Studio)

1. The customer clicks **Custom** on a pricing page.
2. A short brief opens, pre-filled with their organization, current plan, size and branch.
3. A **lead** is created in Core, and you get a WhatsApp/email notification.
4. Studio replies with a discovery call and a quote. If accepted, it becomes a **Studio project**.
5. Delivered custom modules are enabled for that one organization through **entitlements** (feature flags). The product stays one codebase.

9. Money: $0 until the first subscriber

STAGE	TRIGGER	WHAT YOU PAY FOR	MONTHLY
Launch	Now	Nothing. Cloudflare free, Neon free, R2 free, free email tier. (Domain: hemlohq.com, about $12 a year.)	**$0**
First Studio client	A signed quote	Nothing new. Studio work is invoiced and paid, and funds the next step.	**$0**
First subscriber (Systems or School)	First payment	Cloudflare Workers paid (≈ $5) for higher limits; paid messaging only as needed	**≈ $5–10**
~10–20 paying organizations	Revenue covers it	Neon paid (backups, more storage); a PDF rendering service	**≈ $30–60**
Growth	Limits reached	Follow the growth path in `tech-stack-plan.md`: more capacity, a separate branch service if needed, possibly a server or an Iraqi data center	Scales with revenue

Free-tier limits to watch (September 2026):

- Workers: 100,000 requests a day and 10 ms of CPU per request.
- Queues: 10,000 operations a day.
- Cron: 3 schedules.
- R2: 10 GB.
- Neon: 0.5 GB of storage and 100 compute-hours a month.

Upgrade *before* hitting a limit, not after, once revenue allows.

10. Build order

STEP	WHAT	WHY FIRST	COST
1	`hemlohq.com` hub (Arabic + English), with the four branches presented and "notify me" sign-ups	The brand goes live; you can announce	$0
2	`tools.hemlohq.com` with the **launch set of 5 tools**	Brings people in and shows which branch has demand	$0
3	`studio.hemlohq.com` : services, portfolio, quote configurator, brief form	**Earns money soonest** with no infrastructure	$0
4	Hemlo Core: Hemlo ID, organizations, plans, billing, notifications	Needed before any paid product	$0
5	**Systems MVP**, or **School MVP** (whichever Tools and conversations show more demand for)	Deliberately one at a time	$0 until the first subscriber
6	The second product	After the first has paying customers	Paid from revenue

Rule: don't build two products at the same time. Let the free tools and Studio conversations decide which product comes first.

11. Metrics per branch

BRANCH	WHAT TO MEASURE
Hub	Visits → branch clicks; "notify me" sign-ups per branch
Tools	Uses per tool per week; PDFs created; shares; tool → branch sign-up rate
Studio	Briefs received; quote acceptance rate; average project value; care-plan retention; hours per project vs quoted
Systems	Trials started; trial → paid; active locations; monthly churn; job cards per week
School	Schools in trial; students under management; fee collection rate after reminders; renewals per year
Core	Monthly recurring revenue; payment success (FIB/ZainCash/cash); support requests per customer

12. Risks and how the design handles them

RISK	MITIGATION
Four branches spread one person too thin	One product at a time (step 5). Tools are small and static. Studio is capped.
Systems and School face existing competitors	Niche positioning: stock + jobs + warranty for trades; Iraq-native fees and WhatsApp for schools. Validate with 10 conversations each before building.
Studio work crowds out product work	A weekly cap, a monthly review, and only reusable parts flow into the platform
Free-tier limits or price changes	Hono and Postgres are portable. Moving to paid tiers or a €6 server needs no rewrite.
Tools give wrong numbers (customs, salaries)	Clearly labelled estimates, a sources note on each tool, a "last updated" date
Payment collection in cash	Invoices with FIB/ZainCash links, reminders, a grace period, then read-only mode, never data deletion
Data-localization rules	Containers and Postgres move to an Iraqi data center when required

13. Summary

- **Hemlo** is the parent and the hub at `hemlohq.com`.
- **Hemlo Tools** is free forever. It brings businesses in at $0 cost.
- **Hemlo Studio** does custom work at quoted prices, earns money first, and receives every "Custom plan".
- **Hemlo Systems** is one core with an edition per trade, starting with repair workshops (stock, jobs, warranty), on plans.
- **Hemlo School** is complete school management with Iraq-native fees and parent links, on plans.
- **Hemlo Core** is one login, one billing system and one design system for all of them.
- It runs on **Cloudflare + Neon Postgres at $0** until the first subscriber. Then it grows only as revenue allows.